SOLAR POWER SYSTEM

Wiring Guide & Project Documentation

300W Panel | LiFePO4 100Ah Battery | BL912 MPPT Controller | 300W Inverter

1. System Overview

This document describes the wiring, component selection rationale, and operational guidelines for a 12V off-grid solar power system. The system is designed to power both DC loads (DC bulbs directly from battery) and AC loads (AC bulbs via inverter), with water pumping capability using the Lenhard solar pump.

The system follows the standard MPPT solar wiring sequence as described in the BL912 Solar Controller manual:

- Solar panels connect to the charge controller input
- Battery connects to the charge controller battery terminals
- DC loads connect to the charge controller load output terminals
- AC loads connect through the power inverter which draws from the battery
- The water pump connects directly to the battery (12V DC, 180W)

2. Component Specifications

Component	Specification	Key Detail
Solar Panel	300W Monocrystalline	Max input: feeds MPPT controller
Battery	LiFePO4 12.8V 100Ah (Drompower)	1280Wh capacity, 5000+ cycle life
Charge Controller	BL912 MPPT	Auto 12V/24V detection, Bluetooth app
Power Inverter	Drompower 300W DC12V to AC220V	Modified sine wave, USB 5V output
Water Pump	Lenhard LSWQB-12, 180W DC12V	Max head 15M, flow 1.5m³/h

DC Loads	DC bulbs (12V)	Connected via controller load terminals
AC Loads	AC bulbs (220V)	Connected via inverter AC output

3. Wiring Diagram — Step by Step

Step 1: Battery to Charge Controller

Connect the battery to the charge controller FIRST before connecting anything else. This is a critical safety requirement stated in the BL912 manual — connecting in the wrong sequence can damage the controller permanently.

- Battery POSITIVE (+) terminal → Charge controller BAT+ terminal
- Battery NEGATIVE (–) terminal → Charge controller BAT– terminal
- Use thick gauge cable (minimum 6mm² recommended for 100Ah battery) 18AWG<
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> **i** The BL912 controller will auto-detect your 12V battery and set parameters automatically on first connection.

Step 2: Solar Panel to Charge Controller

After the battery is connected, connect the solar panel to the controller's PV input terminals.

- Solar panel POSITIVE (+) → Charge controller PV+ / INPUT+ terminal
- Solar panel NEGATIVE (–) → Charge controller PV– / INPUT– terminal
- Your 300W panel at 12V system will produce roughly 20–25A of charge current — well within the BL912's capacity

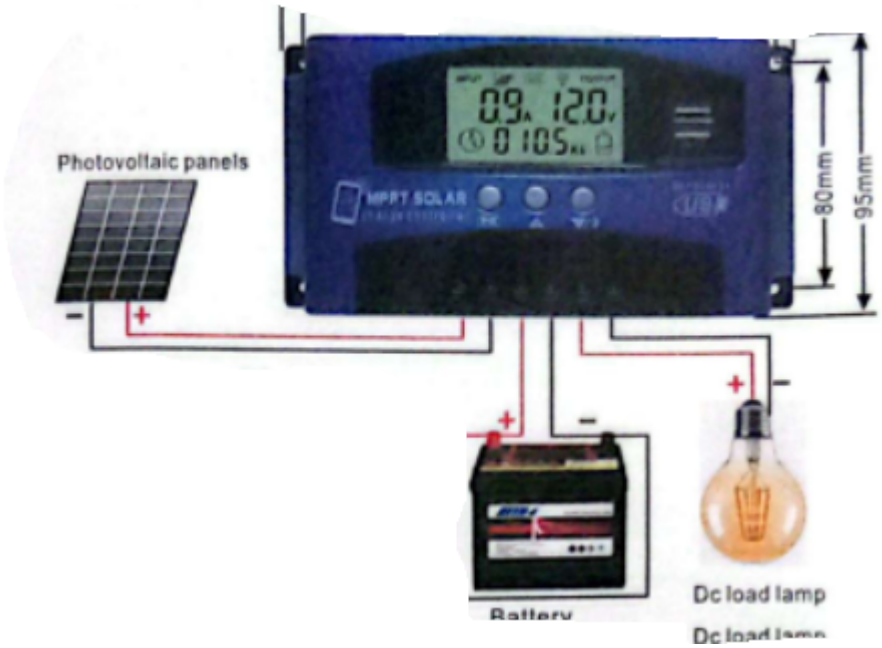
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> ⚠ **Never connect the solar panel before the battery. The controller must see the battery first or it may be damaged.**

Step 3: DC Loads (DC Bulbs) to Charge Controller

The BL912 charge controller has dedicated LOAD output terminals specifically for DC loads. Your DC bulbs connect here. This is the correct and safe way to power DC devices — the controller manages low-voltage cutoff automatically to protect the battery.

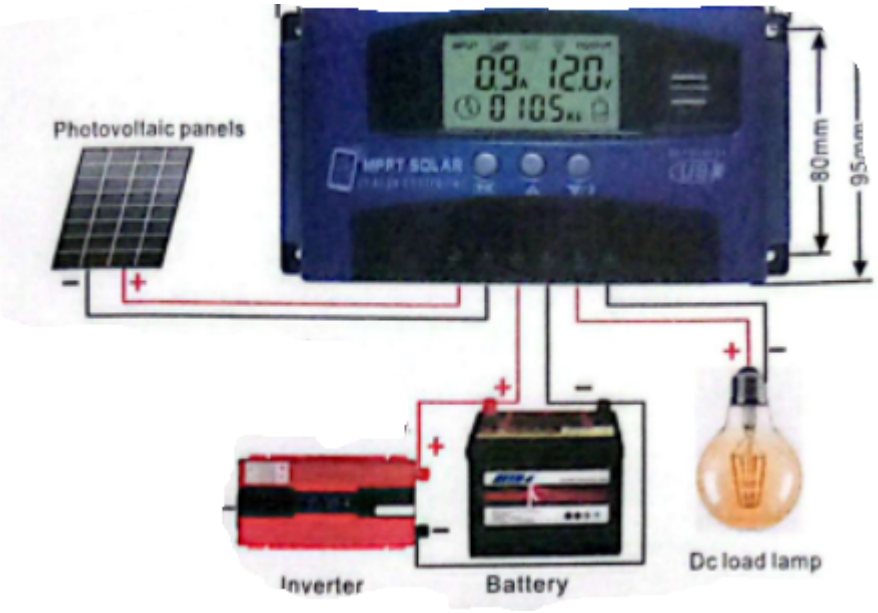
- DC bulb positive wire → Charge controller LOAD+ terminal
- DC bulb negative wire → Charge controller LOAD– terminal
- You can run multiple DC bulbs in parallel from these terminals

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> **i** The controller's load output can be set to automatic (light control) or manual mode. In automatic mode it switches the load on at dusk and off at dawn — useful for lighting.

Step 4: Inverter to Battery (for AC Loads)

The 300W Drompower inverter connects directly to the battery terminals, NOT to the charge controller. The inverter draws high current and needs a direct battery connection for stable operation.

- Inverter V+ (red terminal) → Battery POSITIVE (+)
- Inverter V– (black terminal) → Battery NEGATIVE (–)
- Your AC bulbs then plug into the inverter's AC 220V output socket
- USB devices can use the inverter's 5V USB port
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> ⚠ **The inverter manual states: use thick, short cables between the battery and inverter. Long or thin cables cause voltage drop, overheating, and poor inverter performance. Keep the cable under 1 metre where possible.**

Step 5: Water Pump to Battery

The Lenhard LSWQB-12 pump runs on DC 12V and draws 180W — which is 15A at 12V. It should also connect directly to the battery for adequate current supply.

- Pump red wire (+) → Battery POSITIVE (+) via a 20A fuse
- Pump black wire (–) → Battery NEGATIVE (–)
- Install an inline switch for convenient on/off control

> ⚠ **The pump label says 'FORBID WORKING EMPTY' — never run this pump without water. Running it dry will burn out the motor immediately.**

4. Connection Sequence Summary

Always follow this exact connection order to avoid damaging any components:

Order	Connection	From	To
1st	Battery to Controller	BAT+ / BAT–	Controller BAT terminals
2nd	Solar Panel to Controller	Panel +/–	Controller PV terminals
3rd	DC Bulbs to Controller	Bulb wires	Controller LOAD terminals
4th	Inverter to Battery	Inverter V+/V–	Battery terminals directly
5th	AC Bulbs to Inverter	Bulb plug	Inverter AC socket
6th	Water Pump to Battery	Pump +/– via fuse	Battery terminals directly

5. Why We Chose These Components

LiFePO4 Battery (over Lead Acid)

The Drompower LiFePO4 12.8V 100Ah battery was chosen over conventional lead-acid batteries for several important reasons:

- 5000+ cycle life vs 300–500 cycles for lead-acid — significantly longer lifespan
- Can safely discharge to 80–90% depth without damage. Lead-acid batteries should only discharge 50% to avoid damage
- 1280Wh of usable energy — substantially more usable capacity per kg
- No maintenance required — no water topping up, no sulphation risk
- Safer chemistry — LiFePO4 is the most thermally stable lithium chemistry, no fire risk under normal use

> **i** The BL912 controller manual specifically mentions LiFePO4 (B01) as a supported battery type. Set battery type to B01 in the controller settings for correct charging parameters. The correct charging voltage for LiFePO4 is 14.6V which matches what is printed on your battery label.

MPPT Charge Controller (over PWM)

The BL912 uses MPPT (Maximum Power Point Tracking) technology. This is significantly more efficient than the cheaper PWM controllers:

- MPPT extracts 20–30% more energy from the same solar panel compared to PWM
- Your 300W panel with MPPT will consistently deliver more charge, especially in partial shade or early morning and evening
- The BL912 has a Bluetooth app (ZhiJinPower) for real-time monitoring of charge current, voltage, power generation, and battery status from your phone
- Supports multiple battery types including LiFePO4, gel, and lead-acid

300W Inverter

The 300W Drompower inverter matches the output capacity of your solar panel, which is a sensible balance. The inverter converts 12V DC from your battery into 220V AC for standard household appliances and your AC bulbs.

- Rated at 300W continuous with 600W peak surge — handles the startup surge of most small appliances
- Universal protection circuit covers thermal, overvoltage, undervoltage, overload, short circuit, and earth fault
- Includes a USB 5V port for phone charging without using the AC socket

6. Important Considerations & Safety

Cable Sizing

Using cables that are too thin is one of the most common mistakes in solar installations. Thin cables cause resistance, voltage drop, overheating, and can start fires.

- Battery to controller: minimum 6mm² cable
- Battery to inverter: minimum 10mm² cable (inverter draws up to 25A at full load)
- Battery to pump: minimum 4mm² cable with a 20A inline fuse
- Controller load terminals to DC bulbs: 2.5mm² is sufficient for small DC bulbs

Fusing

- Install a fuse as close to the battery positive terminal as possible on every circuit
- Inverter circuit: 30A fuse
- Water pump circuit: 20A fuse
- The charge controller has built-in protection for the load output circuit

> ⚠ **A battery of this capacity (100Ah LiFePO4) can deliver hundreds of amps in a short circuit. Without fuses, a wiring fault can cause a fire or explosion within seconds. Fuses are not optional.**

Battery Configuration on the Controller

The BL912 controller manual states that when charging lithium batteries, you must set the correct battery type before use. Follow these steps on first setup:

- Power on the controller with the battery connected
- Press MENU and navigate to battery type setting (parameter 6 in the manual — bAt)
- Set to B01 for LiFePO4 lithium iron phosphate
- The controller will then apply the correct 14.6V charge voltage matching your battery specification

Inverter and Battery Protection

The inverter has a battery alarm that activates at 10.6V and shuts down at 10V to prevent over-discharging the battery. However, your LiFePO4 battery has its own BMS (Battery

Management System) that will also cut off discharge at approximately 10V as printed on the label (discharge cut-off 10V ±0.2V). These two protection systems work together.

Water Pump — Critical Note

The Lenhard pump is a surface pump, not submersible. It must be positioned above the water source and can pull water up to 5 metres (max suction head). It can then push water upward a further 15 metres (max head). Always prime the pump before starting and never let it run without water flowing.

Monitoring Your System

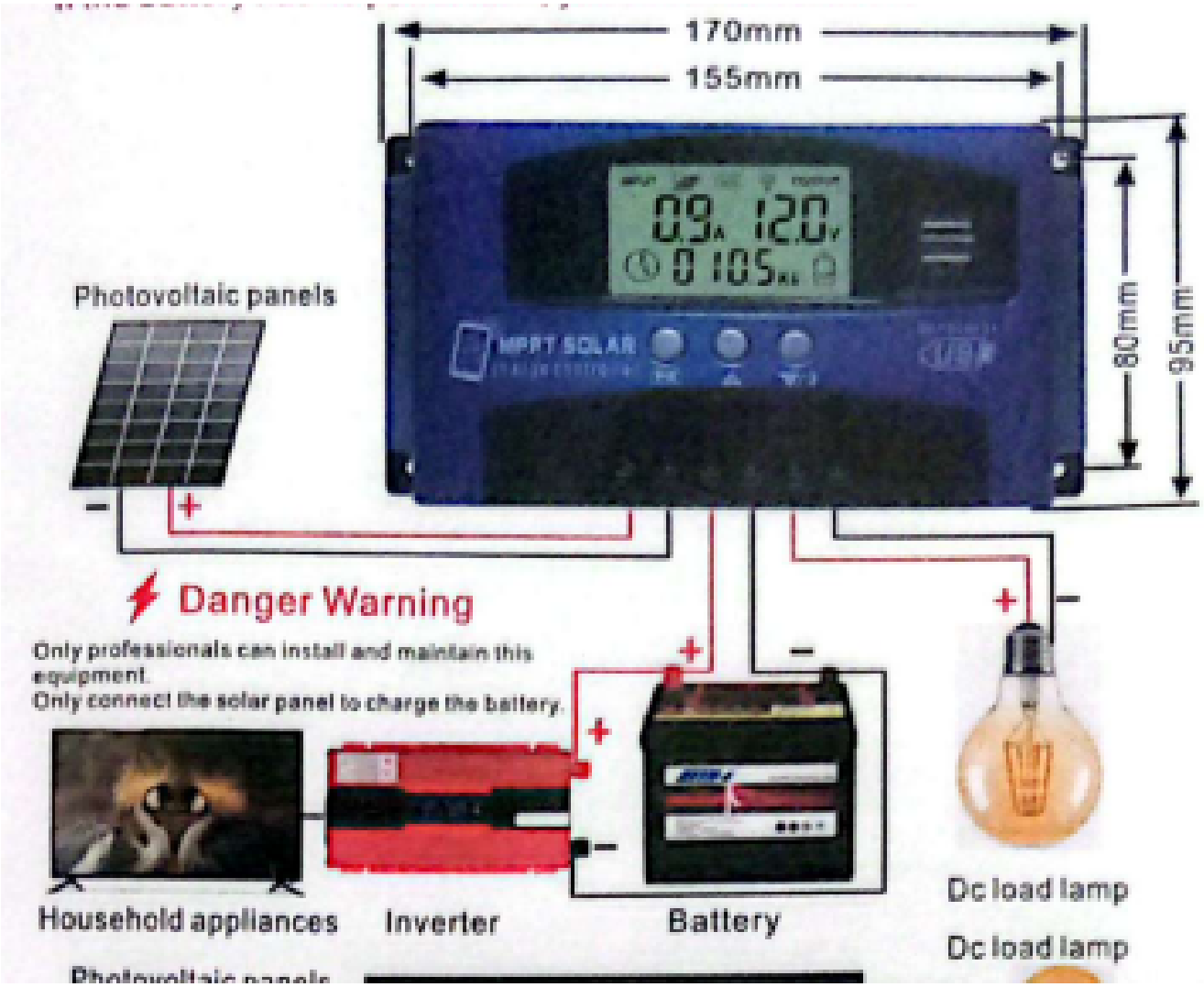
The BL912 controller supports the ZhiJinPower app via Bluetooth. Download it and pair it with your controller to monitor real-time solar input, battery voltage, charge current, daily power generation, and load consumption. This is very useful for understanding how your system is performing and catching any issues early.

7. Estimated Power Budget

Use this as a guide to understand how long your battery will last running your loads:

Load	Power	Hours/Day	Daily Consumption
DC bulbs (e.g. 4 x 10W)	40W	6 hours	240Wh
AC bulbs (e.g. 4 x 15W + inverter loss)	~75W	4 hours	300Wh
Water pump	180W	1 hour	180Wh
Total daily consumption			~720Wh
Battery usable capacity (LiFePO4 90%)			1152Wh
Days of autonomy without sun			~1.6 days
Solar input (300W panel, 5 peak hours)			~1200Wh/day

> **i** Your 300W panel should comfortably cover your daily consumption on a good sunny day in Kenya, with surplus energy going into the battery for evening and night use.


170mm
155mm
80mm
95mm
MPPT SOLAR
Photovoltaic panels
Danger Warning
Only professionals can install and maintain this equipment.
Only connect the solar panel to charge the battery.
Household appliances
Inverter
Battery
Dc load lamp
Dc load lamp